1. Title page: course, topic, student information

Course: Intelligent Decision Systems
Topic: TOPSIS Normalization Study (Task 3)
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2. Short description of the original decision problem

The objective of this task is to select the best modern smartphone among several options based on multiple performance criteria. A typical consumer wants high performance, excellent camera quality, and long-lasting battery life, all whilst keeping the price as low as possible.

- Number of alternatives: 10
- Number of criteria: 4

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3. Decision matrix, criteria types, and weights

Present your original data:

Alternative	Price (USD)	Battery (mAh)	Camera Score	Perf. Score
iPhone 15 Pro	999.0	3274.0	154.0	7192.0
Samsung Galaxy S24 Ultra	1299.0	5000.0	144.0	7249.0
Google Pixel 8 Pro	999.0	5050.0	153.0	4425.0
OnePlus 12	799.0	5400.0	140.0	6800.0
Xiaomi 14 Pro	899.0	4880.0	144.0	6900.0
Sony Xperia 1 V	1199.0	5000.0	136.0	5200.0
Asus Zenfone 10	699.0	4300.0	130.0	5600.0
Motorola Edge 50 Pro	650.0	4500.0	133.0	4800.0
Nothing Phone (2)	599.0	4700.0	135.0	4600.0
Honor Magic 6 Pro	999.0	5600.0	158.0	7100.0

Define: - Criteria types: - Profit (higher is better): Battery (mAh), Camera Score, Perf. Score - Cost (lower is better): Price (USD) - Weights:

Criterion	Weight
Price (USD)	0.35
Battery (mAh)	0.15

Criterion	Weight
Camera Score	0.25
Perf. Score	0.25

Make sure:

$$\sum w_j = 0.35 + 0.15 + 0.25 + 0.25 = 1$$

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4. Description of the 5 normalization methods

- **Vector normalization**: Divides each value by the square root of the sum of squared values of the respective criterion over all alternatives.
- **Min-max normalization**: Rescales the values of each criterion strictly to a [0,1] interval based on the minimum and maximum observed bounds.
- **Max normalization**: Divides each value by the overall maximum value of its respective criterion.
- **Sum normalization**: Divides each value by the total sum of all values for that criterion.
- **Z-score standardization**: Centers values around zero with a standard deviation of 1.

Each method transforms data differently, which may affect the scale, bounding, and relative spacing between values, consequently altering the final TOPSIS results.

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5. TOPSIS results for each method

Method 1: Vector Normalization

Alternative	Score (Ci)	Rank
OnePlus 12	0.7267	1
Asus Zenfone 10	0.6976	2
Nothing Phone (2)	0.6893	3
Motorola Edge 50 Pro	0.6780	4
Xiaomi 14 Pro	0.6165	5
Honor Magic 6 Pro	0.5462	6
iPhone 15 Pro	0.4984	7
Google Pixel 8 Pro	0.4084	8
Samsung Galaxy S24 Ultra	0.3315	9
Sony Xperia 1 V	0.2326	10

Method 2: Min-Max Normalization

Alternative	Score (Ci)	Rank
Honor Magic 6 Pro	0.6686	1
OnePlus 12	0.6533	2
Xiaomi 14 Pro	0.6248	3
iPhone 15 Pro	0.5866	4
Nothing Phone (2)	0.5351	5
Asus Zenfone 10	0.5149	6
Motorola Edge 50 Pro	0.5126	7
Google Pixel 8 Pro	0.4617	8
Samsung Galaxy S24 Ultra	0.4460	9
Sony Xperia 1 V	0.2708	10

Method 3: Max Normalization

Alternative	Score (Ci)	Rank
OnePlus 12	0.7274	1
Asus Zenfone 10	0.6666	2
Nothing Phone (2)	0.6555	3
Motorola Edge 50 Pro	0.6437	4
Xiaomi 14 Pro	0.6266	5
Honor Magic 6 Pro	0.5726	6
iPhone 15 Pro	0.5118	7
Google Pixel 8 Pro	0.4076	8
Samsung Galaxy S24 Ultra	0.3661	9
Sony Xperia 1 V	0.2512	10

Method 4: Sum Normalization

Alternative	Score (Ci)	Rank
OnePlus 12	0.7267	1
Asus Zenfone 10	0.7004	2
Nothing Phone (2)	0.6920	3
Motorola Edge 50 Pro	0.6808	4
Xiaomi 14 Pro	0.6158	5
Honor Magic 6 Pro	0.5440	6
iPhone 15 Pro	0.4976	7
Google Pixel 8 Pro	0.4081	8
Samsung Galaxy S24 Ultra	0.3287	9
Sony Xperia 1 V	0.2307	10

Method 5: Z-Score Standardization

Alternative	Score (Ci)	Rank
Honor Magic 6 Pro	0.6565	1
OnePlus 12	0.6533	2
Xiaomi 14 Pro	0.6109	3
Nothing Phone (2)	0.5690	4
iPhone 15 Pro	0.5435	5
Motorola Edge 50 Pro	0.5409	6
Asus Zenfone 10	0.5269	7
Google Pixel 8 Pro	0.4941	8
Samsung Galaxy S24 Ultra	0.4171	9
Sony Xperia 1 V	0.2923	10

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6. Comparative analysis with tables or plots

Complete Comparative Data Table

Rank	Vector	Min-Max	Max	Sum	Z-Score
1	OnePlus 12 (0.7267)	Honor Magic 6 Pro (0.6686)	OnePlus 12 (0.7274)	OnePlus 12 (0.7267)	Honor Magic 6 Pro (0.6565)
2	Asus Zenfone 10 (0.6976)	OnePlus 12 (0.6533)	Asus Zenfone 10 (0.6666)	Asus Zenfone 10 (0.7004)	OnePlus 12 (0.6533)
3	Nothing Phone (2) (0.6893)	Xiaomi 14 Pro (0.6248)	Nothing Phone (2) (0.6555)	Nothing Phone (2) (0.6920)	Xiaomi 14 Pro (0.6109)
4	Motorola Edge 50 Pro (0.6780)	iPhone 15 Pro (0.5866)	Motorola Edge 50 Pro (0.6437)	Motorola Edge 50 Pro (0.6808)	Nothing Phone (2) (0.5690)
5	Xiaomi 14 Pro (0.6165)	Nothing Phone (2) (0.5351)	Xiaomi 14 Pro (0.6266)	Xiaomi 14 Pro (0.6158)	iPhone 15 Pro (0.5435)
6	Honor Magic 6 Pro (0.5462)	Asus Zenfone 10 (0.5149)	Honor Magic 6 Pro (0.5726)	Honor Magic 6 Pro (0.5440)	Motorola Edge 50 Pro (0.5409)
7	iPhone 15 Pro (0.4984)	Motorola Edge 50 Pro (0.5126)	iPhone 15 Pro (0.5118)	iPhone 15 Pro (0.4976)	Asus Zenfone 10 (0.5269)
8	Google Pixel 8 Pro (0.4084)	Google Pixel 8 Pro (0.4617)	Google Pixel 8 Pro (0.4076)	Google Pixel 8 Pro (0.4081)	Google Pixel 8 Pro (0.4941)

Rank	Vector	Min-Max	Max	Sum	Z-Score
9	Samsung Galaxy S24 Ultra (0.3315)	Samsung Galaxy S24 Ultra (0.4460)	Samsung Galaxy S24 Ultra (0.3661)	Samsung Galaxy S24 Ultra (0.3287)	Samsung Galaxy S24 Ultra (0.4171)
10	Sony Xperia 1 V (0.2326)	Sony Xperia 1 V (0.2708)	Sony Xperia 1 V (0.2512)	Sony Xperia 1 V (0.2307)	Sony Xperia 1 V (0.2923)

Impact on TOPSIS Results

The results demonstrate that the choice of normalization is not merely a scaling formality but a fundamental determinant of the final decision. The rankings bifurcate into two distinct regimes based on the mathematical properties of the chosen transformation method.

6.1 Classification of Normalization Methods

The observed stability in rankings can be explained by categorizing the methods into two primary mathematical families:

1. **Proportional Mapping Methods (Vector, Max, Sum):** These methods exhibit high stability, generating an almost identical hierarchy wherein the OnePlus 12 consistently takes first place. They scale values by dividing them against a fixed global scalar (a vector norm, an absolute maximum, or a sum). Consequently, they preserve the relative geometric proportionality and original scalar ratios inherent to the raw dataset.

2. **Boundary and Statistical Methods (Min-Max, Z-score):** These methods fundamentally disrupt the ranking, shifting the Honor Magic 6 Pro from sixth place to first. Instead of preserving raw proportional ratios, Min-Max maps the data to a strict [0,1] bounding box anchored explicitly by extreme minimum and maximum values, while Z-score standardizes the data by measuring variances around the population mean.

6.2 Geometric Interpretation and Impact on Distance

TOPSIS evaluates alternatives by measuring their Euclidean distance to hypothetical ideal and anti-ideal solutions in an m-dimensional criteria space. - Under **Proportional Methods**, the coordinate space is simply contracted or dilated linearly. Because the original distances between tightly clustered points and outliers are proportionally preserved, the geometric topology remains essentially stable. The OnePlus 12—which maintains mathematically balanced attributes and minimizes cost trade-offs—naturally retains proximity to the positive ideal solution. - Under **Boundary and Statistical Methods**, the geometric space is heavily distorted relative to the data's dispersion. Min-Max normalization violently stretches criteria containing severe outliers. For instance, the $1299 outlier (Samsung) pushes the boundary so far out that a $999 price (Honor Magic 6 Pro) becomes competitively closer to the ideal $599 minimum. Consequently, the Euclidean distance penalty for an expensive device is dampened, allowing the Honor Magic 6 Pro to easily exploit its absolute dominance in the Battery and Camera parameter axes to conquer the distance calculation.

6.3 Deep Analysis of Rank Volatility

The volatility of specific alternatives sheds light on the sensitivity of TOPSIS to preprocessing algorithms: - **Honor Magic 6 Pro (High Volatility, +5 Ranks):** It boasts the dataset's maximum theoretical battery (5600 mAh) and camera specifications but suffers from a high cost. In proportional spaces, this high cost is penalizing relative to the whole vector mass. However, in isolated boundary spaces (Min-Max/Z-Score), it scores a perfect "1.0" or high positive standard deviation on two heavily weighted axes, amplifying its dominance over competitors and catapulting it to 1st place. - **Asus Zenfone 10 (High Volatility, -5 Ranks):** It provides excellent cost-efficiency (low price) but middling technical specifications. Proportional methods logically reward this "jack-of-all-trades" multi-dimensional efficiency. Conversely, statistical standardizations mathematically punish it because its raw specification scores fall firmly below the dataset's arithmetic mean, pushing its geometric coordinate further toward the anti-ideal state.

6.4 Stability Profile

In the context of multi-criteria decision-making, "stability" refers to the algorithm's resistance to emitting drastically different hierarchies when confronted with minor variances or scale alterations in the input matrix. Vector and Sum normalizations prove incredibly durable in this experiment because they dilute the impact of singular extreme values across the global metric sum. By contrast, Min-Max is inherently unstable when criteria contain skewed distributions or outliers, as the entire interval geometry hinges precariously upon only two data points (the absolute minimum and maximum).

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7. Final conclusions

- **Does normalization affect the TOPSIS result?** Yes, without question. Normalization redefines what it means to be "close" to an ideal geometric solution and leads to completely different final recommendations.
- **In which situations does normalization choice become critical?** The choice becomes critical when the dataset's criteria have extreme boundaries or strong outliers. When raw data dispersion behaves unexpectedly, specific normalizations like Min-Max are easily skewed, throwing standard rank stability off balance.
- **Which method is most appropriate for this dataset?** For a consumer product dataset, Vector Normalization serves best. Its proportional mapping rewards well-rounded options (OnePlus 12: excellent price with strong overall specs) compared to finding boundary maximums that highlight inherently disjointed pricing (Honor Magic 6 Pro: high price with top-tier battery/camera).
- **What is the practical implication for decision-making?** Preprocessing data injects implicit theoretical bias. The decision-maker using TOPSIS must actively

decide if the mathematical framework should measure scale proportionality (Vector) or evaluate absolute threshold boundaries (Min-Max).

Final Remark: Normalization is not just a preprocessing step; it can significantly influence the outcome of multi-criteria decision-making methods such as TOPSIS.